Task 2:

Using the MSE loss $l(\beta) = \sum_{i=1}^{N} (y_i^* - X_i\beta)^2$

$= (Y^* - X\beta)^T (Y^* - X\beta)$

and its derivative (with $y_i^* \in \{-1, 1\}$)

$$\frac{\partial l}{\partial \beta} = \frac{\partial}{\partial \beta} (Y^* - X\beta)^T (Y^* - X\beta)$$

$$= \frac{\partial}{\partial \beta} (Y^*)^T Y^* - 2 \frac{\partial}{\partial \beta} (Y^*)^T X\beta + \frac{\partial}{\partial \beta} (X\beta)^T (X\beta)$$

$$= -2X^T Y^* + 2X^T X \beta$$

Setting $\frac{\partial l}{\partial \beta} \overset{!}{=} 0$ yields

$$\boxed{X^T Y^* = X^T X \beta \qquad (4)}$$

Now using $X_i = (X_i - \mu_c) + \mu_c$ for class $c \in \{-1, 1\}$

$$\Rightarrow X^T X = \sum_{i=1}^{N} ((X_i - \mu_c) + \mu_c)^T ((X_i - \mu_c) + \mu_c)$$

$$= \sum_{i: y_i^* = -1} ((X_i - \mu_{-1}) + \mu_{-1})^T ((X_i - \mu_{-1}) + \mu_{-1})$$

$$+ \sum_{i: y_i^* = 1} ((X_i - \mu_1) + \mu_1)^T ((X_i - \mu_1) + \mu_1)$$

$$= \sum_{i: y_i^* = -1} (X_i - \mu_{-1})^T (X_i - \mu_{-1}) + \overbrace{\sum_{i: y_i^* = -1} (X_i - \mu_{-1})^T}^{=0} \mu_{-1}$$

$$+ \mu_{-1}^T \underbrace{\sum_{i: y_i^* = -1} X_i - \mu_{-1}}_{=0} + N_{-1} \mu_{-1}^T \mu_{-1}$$

$$+ \sum_{i: y_i^* = 1} (X_i - \mu_1)^T (X_i - \mu_1) + \overbrace{\sum_{i: y_i^* = 1} (X_i - \mu_1)^T}^{=0} \mu_1$$

$$+ \mu_1^T \underbrace{\sum_{i: y_i^* = 1} X_i - \mu_1}_{=0} + N_1 \mu_1^T \mu_1$$

$$= \overbrace{\sum_{i: y_i^* = -1} (X_i - \mu_{-1})^T (X_i - \mu_{-1}) + \sum_{i: y_i^* = 1} (X_i - \mu_1)^T (X_i - \mu_1)}^{= N\Sigma}$$

$$+ N_{-1} \mu_{-1}^T \mu_{-1} + N_1 \mu_1^T \mu_1$$

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$$\Rightarrow X^T X = N\Sigma + N_{-1}\mu_{-1}^T \mu_{-1} + N_1 \mu_1^T \mu_1$$

Since $N_{-1} = N_1 = \frac{N}{2}$ and $\mu_{-1} + \mu_1 = 0 \Rightarrow \mu_{-1} = -\mu_1$

$$\Rightarrow X^T X = N\Sigma + \frac{N}{2}\left((-\mu_1)^T(-\mu_1) + \mu_1^T \mu_1\right)$$

$$= N\Sigma + N\mu_1^T \mu_1$$

Again, as $\mu_1 = -\mu_1 \Rightarrow \mu_1 - \mu_{-1} = 2\mu_1$, so

$$\mu_1^T \mu_1 = \frac{1}{4}(\mu_1 - \mu_{-1})^T(\mu_1 - \mu_{-1})$$

$$\boxed{\Rightarrow X^T X = N\Sigma + \frac{N}{4}(\mu_1 - \mu_{-1})^T(\mu_1 - \mu_{-1}) \qquad (44)}$$

Substituting (44) into (4) together with

$$X^T y^* = \sum_{i: y_i^* = 1} x_i^T - \sum_{i: y_i^* = -1} x_i^T$$

$$= N_1 \mu_1^T - N_{-1}\mu_{-1}^T$$

$$= \frac{N}{2}(\mu_1 - \mu_{-1})^T$$

$$\Rightarrow X^T y^* = X^T X \beta \Leftrightarrow \left(N\Sigma + \frac{N}{4}(\mu_1 - \mu_{-1})^T(\mu_1 - \mu_{-1})\right)\beta$$

$$= \frac{N}{2}(\mu_1 - \mu_{-1})^T$$

$$\Leftrightarrow \boxed{\Sigma\beta + \frac{1}{4}(\mu_1 - \mu_{-1})^T(\mu_1 - \mu_{-1})\beta = \frac{1}{2}(\mu_1 - \mu_{-1})^T}$$

Set $\tau' := (\mu_1 - \mu_{-1})\beta$, so

$$\Sigma\beta + \frac{1}{4}(\mu_1 - \mu_{-1})^T \tau' = \frac{1}{2}(\mu_1 - \mu_{-1})^T$$

$$\Leftrightarrow \Sigma\beta = \frac{1}{2}(\mu_1 - \mu_{-1})^T - \frac{1}{4}\tau'(\mu_1 - \mu_{-1})^T$$

$$= \left(\frac{1}{2} - \frac{\tau'}{4}\right)(\mu_1 - \mu_{-1}^T)$$

Essentially the provided Solution, they both are very similar

Finally, set $\tau := \left(\frac{1}{2} - \frac{\tau'}{4}\right)$ to get

$$\beta = \tau\Sigma^{-1}(\mu_1 - \mu_{-1})^T$$